Scheduler Equivalence and Performance Trade-offs under a Common LLM-Serving Contract

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Scheduler rankings can conceal a more basic question: do distinct policies actually select different work? We study four large language model serving adaptations under a common execution contract that fixes information, admission, memory reservations, and batch timing. An admission-and-progress invariant proves that chunked first-come, first-served and decode-first select identical per-request work from an empty system, even with binding work budgets. Their normalized transitions and terminal outcomes coincide; the order of entries in an execution journal need not. A separate finite-work certificate rules out activation of the implemented decode-urgency gate in the evaluated fixed-duration regimes. A controlled prompt-ordering ablation identifies the source of the remaining performance trade-offs. Across 120 production-derived windows, the prompt-ordered adaptation reduces aggregate mean latency by 11.3–34.0% relative to chunked first-come, first-served, alongside throughput losses and, on one source, increased mean window-p95 latency. Source-mean preferences exhibit no reversals exceeding the declared 10% margin in both directions, while window-level and cross-metric reversals persist. A composition-dependent timing study preserves the pair's equivalence on a selected subset but exposes window-sensitive mean-latency effects under both clocks. The theorem assumes fixed-duration batches; the empirical results describe a finite, temporally related corpus. The synthetic timing models establish neither hardware performance nor universal scheduler rankings.

CCS Concepts: • **Computing methodologies** → **Simulation evaluation**; • **Software and its engineering** → *Software performance.*

Additional Key Words and Phrases: LLM serving, performance evaluation, scheduling, trace-driven simulation, ranking stability

1 Introduction

Distinct scheduling priority lists can select the same work and induce the same request-state transitions. Aggregate rankings cannot distinguish this structural equivalence from coincidentally similar performance. Evaluation therefore requires establishing what the execution model allows each policy to change.

For large language model (LLM) serving, workload sensitivity, metric dependence, and throughput–latency trade-offs are already well established [1, 4, 20]. Vidur and Vidur-Bench support shared simulation and workload/policy comparisons [3]; evaluation guidelines address baseline fairness and misleading summaries [2]. Those contributions motivate a more specific question: *which nominal scheduler distinctions survive a declared execution contract, why do others disappear, and what performance trade-offs remain?* Answering it requires examining state transitions and mechanism activation alongside performance measurements.

Our methodological approach is to derive effective scheduler behavior from the contract before interpreting performance differences: establish reachable-state equivalence under explicit assumptions, determine whether the implemented mechanisms can activate, and measure the trade-offs that survive. The state-transition argument, inactivity certificate, and saved execution checks below support distinct parts of this analysis; observed agreement outside the proof's assumptions is not a general equivalence theorem.

We investigate four project-maintained adaptations under CM-0.1, a contract specifying online information, legal actions, work advancement, batch timing, and resource acquisition and release. The adaptations retain selected ideas from first-come, first-served (FCFS) serving, chunked prefill, decode-first scheduling, and urgency-aware prompt ordering [4, 7, 17]. The baseline fixes batch cost; a separate sensitivity varies it with executed work. Four or eight

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resident slots preserve admission competition. This abstraction permits a precise account of executable distinctions, without predicting a particular GPU or comparing native serving systems. The complete selected corpus has 120 frozen production-derived windows, six joint arrival/deadline settings, and two residency capacities.

The scientific contribution connects three questions and their answers:

RQ1: Effective behavior. Which policy distinctions change request execution? We prove a resident-prefix invariant: all residents except the youngest collectively demand strictly less than the batch budget. This implies identical selected work for fixed chunked-FCFS and decode-first policies starting empty, including binding budgets. Common batch-end updates then give identical normalized transitions and terminal outcomes (Theorem 3.2). The 1,440 matched terminal comparisons and a separate 72-cell budget sensitivity check implementation conformance; they are not the basis of the universal quantifier within the theorem's scope.

RQ2: Mechanism inactivity. Why does SLO-gated's urgency rule not execute? The service-level-objective (SLO) adaptation uses a fixed token-gap threshold. Clock updates give zero anchor age in the original contract. A finite-work bound also rules out first activation under the examined binding budgets. Completion deadlines do not enter the gate. The measured SLO-gated effects consequently cannot be attributed to urgency; prompt-length admission ordering remains an active distinction. A controlled ablation isolates its latency–throughput–deadline trade-off under the original nonbinding budget, using proven-equivalent existing controls.

RQ3: Residual trade-offs and rankings. What preferences remain, and at what comparison level? Measured mean-latency improvements coexist with throughput losses and, on BurstGPT, larger mean window-p95 latency. Source-mean practical stability coexists with window-level and metric-dependent reversals. Removing duplicate behaviors and endpoints preserves that contrast. A synthetic timing sensitivity retains the chunked/decode-first equivalence while demonstrating that full-corpus mean-latency gains need not hold on a selected window subset.

Section 3 gives the structural arguments; Sections 5 and 6 quantify the surviving comparisons. Event reconstruction and numerical verification support the findings. The separate historical diagnostic in Section 7 examines statistical calibration; it adds no common-contract observations. The selected windows support finite-corpus conclusions. Transfer to other execution contracts remains an open modeling question.

2 Related Work and Evaluation Scope

Simulation and benchmark comparisons. Vidur combines operator profiling and predictive modeling with hierarchical scheduling; its Vidur-Bench suite supplies workloads, scheduling/batching policies, and performance metrics, while Vidur-Search explores deployment configurations [3, Sections 4–6]. Beyond these established comparisons, we ask which policy distinctions the shared model preserves: Section 3 proves reachable-state equivalence for fixed policies, and Section 3.5 bounds mechanism activation. LLMServingSim and its heterogeneous/disaggregated successor also model hardware/software interactions [10, 11]. Our fixed-duration work ledger instead permits direct reasoning about execution semantics. Journal reconstruction verifies conformance to that model; physical prediction would require separate validation [21].

Evaluation methodology and metrics. Papaioannou and Doudali study how datasets and inference scenarios change LLM-system evaluations [20]. Agrawal et al. identify anti-patterns in baseline comparisons, evaluation settings, and metric design, with a speculative-decoding case study [2, Sections 3–4]; Etalon introduces a fluidity index to capture streaming experience missed by conventional summaries [1, Sections 3–5]. We adopt their caution about benchmark and metric choices. Section 6 adds a matched account of source-mean, window-level, and cross-metric preferences after

demonstrated redundancies are removed. Its absence of source-mean practical reversals has a specified threshold and denominator; it does not establish universal portability. We introduce neither a new streaming metric nor a general evaluation checklist.

Serving and scheduling mechanisms. Orca introduces iteration-level scheduling [29]; vLLM's PagedAttention addresses key–value (KV) cache fragmentation and sharing [17]; Sarathi-Serve combines chunked prefill with stall-free batching [4]. Fu et al. learn relative output-length order to approximate shortest-job-first scheduling [13]. Our prompt-length ordering uses observed inputs, without predicting output length or claiming the same objective or optimality. Bari et al. give a formal batching framework and distinguish the throughput-optimal construction from practical SLO-Aware LLM Inference (SLAI), which combines decode urgency and known prompt lengths [7]. SOLA jointly considers request/system state and work allocation to improve SLO attainment [16]. These systems evaluate active state-dependent mechanisms under their own execution assumptions; our adaptation has a fixed, inactive urgency gate. Full-context reservations also omit native paging, while disaggregation and migration, exemplified by DistServe and Llumnix [24, 32], lie outside the evaluated contract.

Length information and application objectives. FastServe uses known input length to initialize a skip-join multilevel feedback queue, with output-token preemption and GPU–host state movement [27]. In contrast, our residents cannot be preempted; known prompt length changes only waiting admission order in SLO-gated. PLA-Serve, called LAPS in its published PDF, separates long and short prefills and combines queue separation with batching and execution optimizations [22]. Its prompt-length awareness is therefore not equivalent to sorting one waiting list under constant batch cost. These systems optimize measured serving performance; our question is which differences remain executable after the contract is fixed.

JITServe uses imprecise request information refined during generation to allocate service toward heterogeneous application objectives and maximize goodput [31]. Its active objective-aware decisions contrast with our hidden metric deadlines and fixed token-gap gate. Our inactivity certificate concerns this particular adaptation, not a failure of objective-aware scheduling in JITServe or SLAI. Comparing their reported gains directly with ours would conflate information, resource, and execution assumptions.

Binding constraints and throughput. Dai et al.'s May 18, 2026 revision studies queueing stability with token-load-dependent batch time and a maximal-request-batch-size extension [12]. The binding constraint matters in both studies, but the estimands differ. Our contract fixes batch time, limits residency to four or eight requests, and measures output divided by finite-window completion duration. Its original-contract chunked work cannot exhaust the token budget; Section 5.7 separately tests lower budgets. These drained rates are not a stability region, and the observed throughput losses neither refute Dai et al.'s results nor establish a new optimality theorem.

Ao et al. model endogenous memory growth and linear iteration times, using a fluid benchmark to derive WAIT for known output lengths and Nested WAIT for unknown lengths [6]. Their stability and asymptotic results are conditioned on memory assumptions; evaluation includes profiled simulation and supplemental GPU evidence. Our full-context reservations instead remove memory-growth admission risk, and our theorem characterizes behavioral equality rather than throughput optimality. The finite drained rates reported here cannot establish or refute those stability claims.

Workload scope. BurstGPT and ServeGen characterize production-derived workloads [26, 28]; resampling has a longer history in scheduler evaluation [30]. Our analysis instead conditions on a fixed selected corpus, with equal-window

source means and explicit matched comparisons. This supplies evidence within those windows, not a new workload generator or an estimate of production-wide scheduler preferences.

Inference and evidence reuse. STAT01 diagnoses established inference procedures [8, 9, 14, 15]; it introduces no new method. Historical inputs and rule outputs are reused, with separate common-model execution identities (Appendix B). Shared infrastructure does not constitute independent validation.

3 A Common Execution Contract

3.1 Why the comparison object matters

The historical corpus contains 9,360 configurations and two deterministic verification passes per configuration. Its primary panel contains eleven policies, with two additional style approximations. The historical engine permits policy-dependent service modes: instantaneous prefill for some policies, explicit prefill work for others, and differing enforcement of KV cache limits. A policy's requested work can also differ from the engine's realized progress. Consequently, those outcomes cannot be interpreted as a controlled comparison of scheduling order. Table 1 distinguishes the principal evidence objects used in this paper; the budget, timing, and prompt-ordering studies are controlled extensions of C01-FULL.

The common contract distinguishes changes in the order of work that all executes from changes in which requests enter or progress. Fixing batch cost removes composition-dependent execution cost; bounded residency retains admission competition. We do not claim prospective selection of these settings for the equivalence or trade-offs reported here.

3.2 Service, resources, and information

CM-0.1 represents one abstract device. Each batch lasts 1 ms and has a 32,768-token work budget. Each executed prompt token consumes one unit; each selected decode request consumes one unit and emits one token. All policies share these rules. A request emits no output during its own prefill, including its final chunk; its decoding starts in a later batch. Other resident requests may decode in the same batch. Baseline batch duration is independent of composition; Section 5.8 varies this rule. All reported seconds and tokens per second are model quantities, not estimates for a particular GPU or LLM.

Each resident request reserves the complete 32,768-token context bound. All frozen request lengths fit; the maximum prompt-plus-output length is 30,053. Capacities of 131,072 and 262,144 tokens therefore permit *four and eight resident requests*, respectively, even though the configured sequence limit is 64. Actual used KV follows executed prompt and output prefixes, rounded to 16-token blocks, and must remain within the reservation. Completion releases the reservation. Waiting requests own none. This conservative rule avoids output-length prediction and allocation deadlock; it does not model incremental native paging or cache eviction.

There is no preemption, dropping, migration, prefix sharing, or recomputation. Requests arriving during a batch become eligible at its end. Idle time advances to the next arrival. The administrative horizon is the final arrival plus 50 s; a batch cannot finish beyond it. Unfinished requests remain explicitly unfinished rather than contributing observed completion latencies.

Policies see only arrived request identifiers, stable order, arrival time, prompt length, acknowledged progress, residency, last-output/prefill-completion anchor, and current time. Actual remaining output length, future arrivals, source identity, and output-derived metric deadlines are hidden. The engine either executes the exact proposed token

Table 1. Evidence objects and the conclusions they support. Historical identifiers containing "faithful" are immutable names, not validation certificates.

Object	Comparison or check	Interpretation
Historical corpus	Policy-dependent service/resource modes; archived ranking arithmetic	Motivation and historical-rule counts
C01-FULL	Four adaptations; common CM-0.1 execution and information	Descriptive finite-corpus performance
STAT01	Synthetic independent studies with known population truth	Scenario-specific inference calibration

Table 2. Project-maintained adaptations. Artifact IDs append `_c01` to `whole_fcfs`, `chunked_fcfs`, `decode_first`, and `slo_gated`. All use the shared contract above.

Adaptation	Historical lineage	Executed ordering rule and principal restriction
Whole-FCFS	vLLM-inspired whole-prompt implementation	Admit whole prompts in arrival order when feasible; otherwise decode residents. Admitting prefills defers all decodes that batch. No native paged allocator.
Chunked-FCFS	vLLM-inspired chunked implementation	Resident work before waiting work in stable order; at most 512 prefill tokens per request per batch.
Decode-first	Sarathi-inspired implementation	Resident decodes, then resident prefills, then waiting prefills; 512-token cap. No original engine timing.
SLO-gated	SLAI-inspired implementation	Urgent decodes first; then resident prefills, waiting prefills by prompt length, and residual decodes. Fixed 0.495 s gate; 512-token cap. No dynamic offset, heterogeneous classes, or full SLAI.

spans or rejects the plan. The shared contract therefore fixes both the resources and the online information available to each adaptation.

3.3 Adaptation identities and effective behavior (RQ1)

Table 2 defines the four evaluated adaptations. Published systems are lineage references; historical project implementations and CM-0.1 adaptations are separate objects. The panel was targeted using the historical comparison questions, not sampled to represent all schedulers.

3.4 Reachable-state equivalence

We distinguish four observations: candidate ordering; selected per-request work; state transitions and timestamps; and terminal outcomes and metrics. Equality at a later level alone does not imply equality at an earlier one. A *normalized journal* sorts the selected and executed spans by request identifier, retaining their types, endpoints, and all other fields.

Assumptions and state. Consider a finite valid trace with positive integer prompt and output lengths p_i, o_i, nondecreasing finite arrivals, and stable identifiers resolving simultaneous arrivals. Each prompt satisfies $p_i \le B$ and each total length fits the reserved context. Both runs start empty and use one fixed policy throughout, the same fixed positive integer budget B, chunk cap $C = 512$, unit decode work, and reservation capacity $K \ge 1$. Admission occurs only on first positive prefill work; slots are released after batch completion. Residents precede waiting requests, admissions use a stable arrival-order prefix, and resident prefills retain that order. There is no preemption, bypass, or external admission.

Exogenous arrivals during a batch become eligible at the next boundary. Batch duration is the same fixed positive value in both runs, with a finite, advancing clock and common horizon.

At a boundary, a request has prefilled progress x_i, generated progress y_i, residency, an anchor, and first-output/completion timestamps. A resident is unfinished and either prefilling or decode-ready. Its work offer before batch-budget truncation is

$$q_i = \begin{cases} \min(C, p_i - x_i), & x_i < p_i, \\ 1, & x_i = p_i. \end{cases} \tag{1}$$

An executable span records request, phase, and progress endpoints; a request appears at most once per batch. Chunked-FCFS visits residents in stable order, then waiting requests. Decode-first visits resident decodes, resident prefills, then waiting requests, preserving stable order within each group. Both allocate the minimum of an offer in Equation 1 and the remaining budget, subject to reservation capacity for new admissions.

Lemma 3.1 (Resident-prefix slack). *At every reachable boundary, admissions have followed a stable arrival-order prefix. For residents $R = (r_1, \ldots, r_m)$ in that order, if $m > 0$,*

$$\sum_{j<m} q_{r_j} \le B - 1. \tag{2}$$

If total resident demand exceeds B, the youngest resident r_m is prefilling.

Proof. The empty initial state satisfies the claim. We give the conditional selection and preservation steps together to avoid assuming the invariant at an unproved successor. Under Equation 2, the earlier residents' offers fit and leave at least one token for the last resident. If total demand fits, both policies serve every offer fully. If it exceeds B, the last resident cannot be a unit-cost decode; it is a prefill and remains last under decode-first. Permuting the earlier residents therefore leaves exactly the same residual allocation to it. Both policies next admit the same waiting prefix with the same residual budget and slots.

Use the resulting chunked-FCFS execution list Q, ordered by stable arrival. Each selected request receives an integer allocation $e_i \ge 1$, with total at most B. Only the final entry can receive less than its offer: a truncated prefill exhausts the budget. A fully served survivor has next offer $q'_i \le e_i$: remaining prefill decreases, a completed prefill becomes a unit-cost decode, and a continuing decode still costs one. A truncated final prefill survives and is the youngest resident; no admission follows it. If any resident survives, let z be the youngest. All other survivors were fully served, including when completions leave holes in the resident order. Consequently,

$$\sum_{i \in R' \setminus \{z\}} q'_i \le \sum_{i \in R' \setminus \{z\}} e_i \le \sum_{i \in Q} e_i - e_z \le B - 1. \tag{3}$$

Equation 3 proves the successor inequality; if z were a decode, its unit offer would make total demand at most B. Empty successor states are immediate. Prefix admission and completion removal preserve arrival order; later arrivals cannot introduce an older identifier. Idle jumps and new waiting arrivals change no resident offer. Induction proves the claim, including simultaneous arrivals and partially processed prefills. □

Theorem 3.2 (Fixed-policy equivalence from an empty system). *Under the assumptions above, chunked-FCFS and decode-first select identical per-request executable spans at every boundary. They have identical normalized journals, request states, timestamps, terminal outcomes, and metrics, for every valid trace and every fixed admissible B and K.*

Proof. Couple the runs at the empty state. The conditional selection argument in Lemma 3.1 gives the same spans at each coupled boundary. Per-request updates commute and share one batch-end timestamp; reservation sums and completion sets are order-independent. Thus successor states, anchors, first outputs, and completions coincide. The lemma preserves the induction hypothesis. Identical idle jumps, arrival eligibility, horizon checks, and stopping decisions also couple any unfinished outcomes. Applying the same metric definitions to identical terminal records proves the final assertion. Ordered span arrays may still differ. □

Binding example and scope. For simultaneous arrivals with prompts $(1024, 1, 1024)$, outputs $(3, 3, 3)$, $B = 1024$, and $K = 4$, the first batch allocates prefills $(512, 1, 511)$. The next offers are $(512, 1, 512)$, totaling $1025 > B$: a later ready decode coexists with an earlier unfinished prefill. Both policies allocate $(512, 1, 511)$, moving only the decode's position in the span list. Thus the theorem does not assume that all offers fit. It covers four/eight residents and budgets 1,024/2,048 when inputs satisfy the unchanged domain. It does not cover arbitrary injected states, policy switching, changing budgets/chunk rules, multi-token decode, waiting-before-resident policies, or asynchronous/preemptive systems. The invariant encloses reachable states; it is not a necessary-and-sufficient characterization of every possible state. Native serving systems are outside this theorem.

Nonbinding corollary and remaining mechanisms. In the original configuration, at most K existing or newly admitted requests each offer at most 512 tokens, so $512K \leq 4096 < 32768$. This simpler sufficient condition also proves equivalence; the theorem explains why reducing the budget alone cannot separate the fixed pair. All 1,440 matched terminal comparisons confirm its realization in the selected corpus. The proof, rather than these finite checks, establishes its scope over valid arrival histories.

The same work bound includes SLO-gated's residual decodes, even though they follow waiting prefills. A newly decode-ready resident's anchor is its prefill batch's end; every later decode resets it to the next batch end. There is no idle jump while such a resident exists. At each scheduling boundary its anchor age is therefore zero under the implemented clock updates, far below 0.495 s. The urgency gate never binds. Prompt-length ordering of waiting admissions remains active; resident prefills still progress, and later arrivals cannot preempt admitted requests. Whole-FCFS also retains a distinct rule: admitting whole prefills defers all decodes in that batch. Neither mechanism is removed by the work bound. Thus the model does more than give policies identical settings: it identifies which of their decision rules can change state transitions. The resulting finite-window completion distributions are the empirical object of Section 5.

3.5 Urgency inactivity and its sufficient condition (RQ2)

Urgency concerns a token gap, not a synthetic completion deadline. At a planning boundary t, an arrived, unfinished resident is critical exactly when its prefill is complete, its anchor α_i is defined, and $t - \alpha_i \geq 0.495$ s. The anchor is the end of final prefill or the last executed decode; an omitted decode does not reset it. Waiting requests and unfinished prefills cannot qualify. Neither completion deadlines nor estimated remaining service enters this predicate. The protocol interprets 0.495 s as a uniform 0.5 s token-gap allowance minus a 5 ms margin, but the implementation fixes the threshold: it exposes no SLO parameter. Changing finite metric deadlines alone cannot change execution.

Finite-work inactivity certificate. A finite-work argument also limits activation when the budget binds. Let P be total prompt work, $B > K - 1$ the integer budget, K the residency limit, and δ the fixed batch duration. Suppose a first critical decision exists. In each consecutive batch omitting that ready decode since its anchor, the residual loop must have exhausted B; at most $K - 1$ other decodes consume one token each. Every such batch therefore executes at least

$B-(K-1)$ prefill tokens. With no recomputation or idle jump while the decode remains resident, m omitted batches imply

$$m[B-(K-1)] \le P, \qquad t-\alpha_i \le \delta \left\lfloor \frac{P}{B-(K-1)} \right\rfloor . \tag{4}$$

By Equation 4, $\delta\lfloor P/[B-(K-1)]\rfloor < 0.495\,\mathrm{s}$ rules out first activation: the finite prompt work cannot sustain enough consecutive decode omissions to reach the urgency threshold. The common-domain restriction gives $P \le NB$ for N requests. For $N = 200$, $B \ge 1,024$, $K \le 8$ and $\delta = 1\,\mathrm{ms}$, the bound is at most 201 ms. This is a sufficient inactivity certificate for these assumptions, not a general waiting-time guarantee or a necessary condition for activation. The original nonbinding case has the tighter zero-age proof above.

The fixed-threshold code agrees with its declared adaptation; configurable token-gap classes in the historical scheduler are a different implementation. Engineering checks at the adjacent floating-point values below and above 0.495, and at equality, confirm the predicate without establishing that such supplied states are reachable. The hidden metric deadline cannot exercise the gate; altering the policy literal would instead change the comparison object.

3.6 Conformance evidence

The event referee reconstructs request states, token spans, timing, completion, and resource use from the inputs and execution journal without importing the engine or policy implementation. It checks that intended and executed work agree and recomputes the metrics. Every campaign cell passed this check; an independent analyzer also checked the complete ranking and reversal reports. The final evidence review additionally checked all result identities and terminal metrics and replayed 24 structurally selected new-cell journals. Seven previously validated hand-derived fixtures cover mixed binding work, completion holes, staggered arrivals, unit budget, and horizon censoring. Checks of 36 saved budget-study journals examine 37,034 boundaries and agree with Lemma 3.1; these finite checks support implementation conformance, not the theorem's universal quantifier. The authoritative engine and policies are unchanged. These checks establish conformance to CM-0.1, not GPU fidelity. Figure 1 summarizes the evidence chain.

AI assistance in the research workflow. OpenAI Codex assisted with software development for the research, experimental analysis and validation, and figure and table preparation. The associated computational workflow includes scripts that check saved input fingerprints, compute performance and ranking summaries, and render displays from frozen numerical evidence. Validation scripts compare reported quantities and supplementary endpoint rows with saved records; separate checks reconstruct request execution and the worked example from event journals (Appendix B). Codex also assisted with packaging and checking the saved-evidence reproduction materials. OpenAI ChatGPT assisted with scientific brainstorming. These are research-assistance roles; writing assistance is reported separately in the declarations. Tool assistance does not itself validate a scientific claim. The author reviewed the resulting work and retains responsibility for the methods, evidence, interpretations, and conclusions.

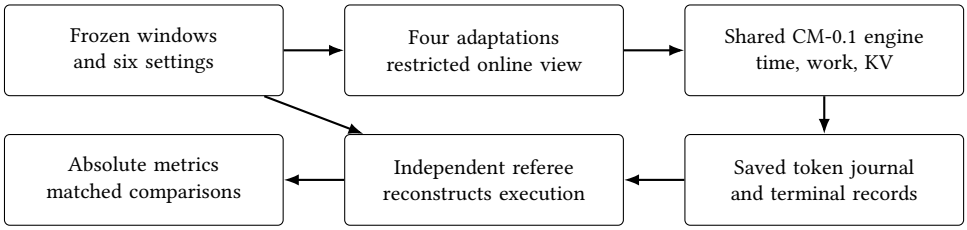


Fig. 1. Comparison and verification under one model. The independent analysis check recomputes aggregates, ranks, and reversals from the absolute metrics. No stage in this chain establishes physical-system accuracy.

4 Corpus, Conditions, and Analysis

4.1 Trace and implementation provenance

The corpus comprises 40 windows of 200 requests from each of Azure 2024, Bailian/Qwen, and BurstGPT [5, 19, 26]. The Azure release identifies DynamoLLM as its accompanying publication [23]; the pinned Bailian/Qwen release requests attribution to the KVCache workload study [25]. These are three separately released sources, not three independent provider populations: BurstGPT and Azure have Azure infrastructure in their collection provenance. Table 3 identifies the consumed files and transformations. The supplement records full file and window fingerprints. The official Azure conversation release has the name `AzureLLMInferenceTrace_conv_1week.csv`; its published SHA-256 matches the consumed file despite the local rename. Azure and BurstGPT are licensed CC BY 4.0, and the pinned Bailian/Qwen data Apache 2.0. Input acquisition instructions retain attribution, license texts, and transformation notices.

Each source's 40-window panel contains ten inherited pilot windows and 30 subsequent windows selected using fixed seeds and valid-row ranges. The expanded manifest was frozen before its corresponding campaign outcomes; earlier pilot outcomes already existed. Within the C01 manifest, row-overlap checks find no overlapping selected ranges. Exact overlap with some related projects' older BurstGPT selections remains unresolved. Nonoverlap within this panel does not establish temporal independence or representativeness. The frozen chronology labels allocate 20/10/10 windows per source to EARLY/MIDDLE/LATE. Four consecutive ten-window blocks provide an additional descriptive sensitivity; their elapsed time spans are unequal.

C01 reuses the historical input reconstruction, while its execution engine, policy views, and adaptation implementations have separate identities. C01-FULL retains C01 execution semantics at commit `9aed0206de66`; its full-campaign execution commit is `2d8342e699f3`. The complete matrix has $120 \times 6 \times 4 \times 2 = 5,760$ cells. Of these, 144 were reused from the pilot only after matching input, code, configuration, and journal identities; 5,616 were newly executed in the completed full campaign. Reuse is not an independent replication. No cell is omitted, and a deterministic cell is not rerun to manufacture a sampling distribution.

4.2 Inherited settings jointly transform arrivals and deadlines

The six labels LOW, PRE_KNEE, KNEE, POST_KNEE, OVERLOAD, and HIGH_PRESSURE come from historical calibration, not from a saturation analysis of CM-0.1. For a window-specific compression factor f, rebased arrival gaps and synthetic deadline slack are both divided by f. Equation 5 expresses the joint transformation:

$$a'_i = (a_i - a_1)/f, \qquad d'_i = a'_i + (d_i - a_i)/f. \tag{5}$$

Table 3. Consumed dataset provenance. Each row contributes 40 windows and 8,000 request records. Hashes are prefixes of SHA-256 values; full values are in the bundled input manifest. The artifact provenance audit matches these hashes to upstream release assets or a pinned LFS object and retains the corresponding license notices.

Source	File and representation	File hash prefix
Azure 2024	`AzureLLMInferenceTrace_conv_2024.csv`; source arrival and token counts	`a0cc9b969a9b`
Bailian/Qwen	`qwen_traceB_blksz_16.jsonl`; input/output lengths read directly; 16-token prefix-block hashes are separate fields	`68e3f98e2d60`
BurstGPT	`BurstGPT_without_fails_2.csv`; retained valid source rows and token counts	`56193aa9b2bb`

The implementation accumulates scaled nonnegative gaps in arrival order. Prompt/output lengths and priorities are unchanged. Factors range from 22.525 to 180.873 for Azure, 17.956 to 186.580 for Bailian, and 102.884 to 725.020 for BurstGPT. Equal labels therefore do not mean equal absolute arrival rates across windows or sources.

Deadlines are synthetic, derived in the inherited workload construction using output-length information; priorities are uniform. These attributes are not observed production service contracts. Deadlines are used only by the metric layer and are hidden from the scheduling adaptations. The setting axis is consequently *joint arrival/deadline stress*. A change in goodput across this axis cannot be attributed solely to arrival load.

4.3 Endpoints and aggregation

For arrival a_i, first output u_i, completion c_i, and output count o_i, latency is $c_i - a_i$ and time to first token (TTFT) is $u_i - a_i$. Time per output token (TPOT) is $(c_i - u_i)/(o_i - 1)$ for $o_i > 1$. The saved means and p95s use their defined completed-request populations; one-output requests do not have TPOT. The p95 uses the saved linear interpolation convention. Completion fraction is completed requests divided by all arrivals. Token throughput is completed output tokens divided by elapsed time from first arrival to final completion, or to the horizon if unfinished. Emitted-token throughput also counts any output of unfinished requests. Request throughput uses the same time denominator. These are finite-window completion rates, including the drain period; they do not estimate a maximum sustainable arrival rate or system capacity.

Arrival-normalized weighted goodput (ANWG) is defined by Equation 6:

$$\mathrm{ANWG} = \frac{\sum_i w_i \, \mathbf{1}\{i \text{ completes and } c_i \le d'_i\}}{\sum_i w_i}. \tag{6}$$

Here $w_i = 1$, so ANWG is the fraction of arrivals meeting their synthetic deadline. It is dimensionless, unlike requests or tokens per second.

The eleven frozen ranking endpoints are ANWG, completion fraction, three throughput measures, and mean/p95 latency, TTFT, and TPOT. Larger is better for goodput, completion, and throughput; smaller is better for the six time endpoints. The complete endpoint set remains in the supplement, including redundant and numerically uninformative endpoints.

The window is the principal analysis unit. At a fixed source, setting, capacity, and policy, source means give each of the 40 windows equal weight. For the compact source–capacity table, we first average the six settings within each window and then average the 40 windows. A mean of window p95s is always labeled as such; it is not a pooled-request p95. Matched differences hold window, setting, and capacity fixed. We show all windows, not a selected favorable subset.

4.4 Ranking and practical-reversal definitions

Exact equality is primary: all maximizers or minimizers enter the top set, and tied policies receive average ranks. The frozen sensitivity admits values within $10^{-12}\max(1, |v_{\text{best}}|)$ of the best; it does not replace exact rankings or the practical-margin rule.

For policies A, B and condition c, let $\Delta_c = q_m(v_{A,c} - v_{B,c})$, where $q_m = 1$ for a larger-is-better endpoint and $q_m = -1$ otherwise. The relative margin in Equation 7 uses the losing value:

$$r_c = \frac{|v_{A,c} - v_{B,c}|}{|v_{\text{loser},c}|}. \tag{7}$$

A practical reversal between c and c' requires $\Delta_c \Delta_{c'} < 0$ and both $r_c > 0.10$ and $r_{c'} > 0.10$. The strict boundary is evaluated exactly on the represented binary values. Zero losing values and missing values are unestimable. Nonzero ties remain eligible with margin zero. All denominators retain these distinctions.

Separate summaries compare within-source window pairs, Cartesian cross-source window pairs, settings within a window, capacities at a fixed window and setting, metrics at a fixed window, setting, and capacity, and source means at a fixed setting/capacity/metric. Cross-source windows are not paired by ordinal. For metric comparisons, each endpoint is oriented separately. These overlapping comparisons describe the finite corpus. We attach no population confidence intervals, p-values, or significance stars to C01.

5 Absolute Performance and Trade-offs (RQ3)

5.1 Completion and performance magnitudes

Every cell completes all 200 requests: 1,152,000 request executions across repeated conditions, not independent requests. There is no horizon censoring and no undefined cell-level latency, TTFT, or TPOT endpoint. TPOT populations range from 103 to 200 because single-output requests are excluded from that particular metric. Across cells, mean latency ranges from 0.008314 to 3.893923 s, p95 latency from 0.008560 to 7.300052 s, and token throughput from 332.489 to 7,766.685 tokens per simulated second.

Table 4 reports absolute source–capacity aggregates. Chunked-FCFS and decode-first share a row because their terminal records and all metrics are exactly equal; their separate identities remain in the artifact.

5.2 Mean latency gains do not imply a universal improvement

SLO-gated reduces aggregate mean latency against chunked-FCFS in all six source–capacity groups, by 11.3–34.0%. The largest aggregate reduction is Azure at four slots, from 2.8132 to 1.8559 s. At eight slots, BurstGPT changes from 0.1907 to 0.1692 s, the smallest percentage improvement. This is a substantial difference in the model's mean response time, but it comes with a throughput reduction in every group, ranging from 1.65% to 4.01%.

The tail comparison is less favorable. For BurstGPT, mean window-p95 latency rises from 1.0005 to 1.0377 s at four slots and from 0.3707 to 0.4410 s at eight slots: increases of 3.72% and 18.97%. Shorter average latency can therefore coexist with longer upper-tail latency under the same contract. At four-slot Bailian, ANWG is also 3.12% lower for SLO-gated despite its 24.05% mean-latency reduction. Throughput, deadline attainment, and response time reward different aspects of the completed schedule.

For a matched, fully completed 200-request condition, mean latency is $\sum_i (c_i - a_i)/200$, whereas token throughput is $\sum_i o_i / (\max_i c_i - \min_i a_i)$. Reducing the sum need not reduce the last completion or an upper quantile. Deadline

Table 4. Absolute performance, equally averaging six settings within each window and 40 windows within each source. "Slots" are effective full-context residents. Mean and p95 columns are mean latency and *mean window-level p95 latency*; throughput is output tokens per simulated second.

Source	Slots	Adaptation	ANWG	Mean (s)	p95 (s)	Tokens/s
Azure 2024	4	Chunked / Decode	0.1746	2.8132	5.3232	3746.2
		Whole-FCFS	0.1746	2.8145	5.3275	3743.5
		SLO-gated	0.2001	1.8559	5.1342	3684.4
Azure 2024	8	Chunked / Decode	0.3174	1.3626	2.5986	7018.7
		Whole-FCFS	0.3101	1.4092	2.6892	6816.5
		SLO-gated	0.3892	0.9357	2.5131	6825.8
Bailian/Qwen	4	Chunked / Decode	0.1828	2.2655	4.2969	3603.2
		Whole-FCFS	0.1799	2.3020	4.3705	3548.3
		SLO-gated	0.1771	1.7205	3.9681	3531.4
Bailian/Qwen	8	Chunked / Decode	0.3330	1.0417	2.0227	6183.4
		Whole-FCFS	0.3189	1.1063	2.1463	5914.1
		SLO-gated	0.3693	0.8087	1.9037	5935.7
BurstGPT	4	Chunked / Decode	0.1030	0.5225	1.0005	3354.9
		Whole-FCFS	0.0889	0.5654	1.0843	3054.9
		SLO-gated	0.2321	0.4448	1.0377	3223.4
BurstGPT	8	Chunked / Decode	0.3072	0.1907	0.3707	5532.1
		Whole-FCFS	0.2129	0.2314	0.4505	4796.5
		SLO-gated	0.4613	0.1692	0.4410	5322.3

attainment counts which completions cross synthetic deadlines; its Bailian loss is therefore compatible with the mean-latency gain. The source/capacity throughput means average window rates and are not inverses of mean durations. The following saved-record illustration connects these distinct objectives to the active admission ordering identified in Section 3.

5.3 Controlled prompt-ordering ablation

ABL01 replaces only SLO-gated's prompt-length sorting of waiting requests with FCFS order, retaining its urgency, resident-prefill, and residual-decode logic. Under the original nonbinding budget, all resident work fits and waiting admissions cannot displace reservations. The control and chunked-FCFS consequently select the same work and share each batch end; induction from the empty state gives normalized trajectory equivalence. This conditional coupling permits reuse of 1,440 existing controls, not new simulations or independent replication of the C01 contrasts.

Removing prompt ordering increases equally weighted aggregate mean latency from 0.9891 to 1.3660 s (38.10%), raises token throughput by 3.21%, and reduces deadline attainment by 6.85 percentage points. Percent changes use the original SLO-gated aggregate as denominator; all 200 requests complete in every pair. Both arms have zero urgency activations. The intervention therefore attributes these trade-offs to waiting-request prompt ordering within this deterministic contract, rather than to urgency. Effects vary: in BurstGPT the control improves mean latency in 211 pairs, worsens it in 198, and ties in 71 under the exact rule. Six nominally nonzero mean-latency differences across the full panel are within 10^{-12} s; TPOT differences of order 10^{-16} s are roundoff. Prompt ordering is neither uniformly beneficial nor established as optimal.

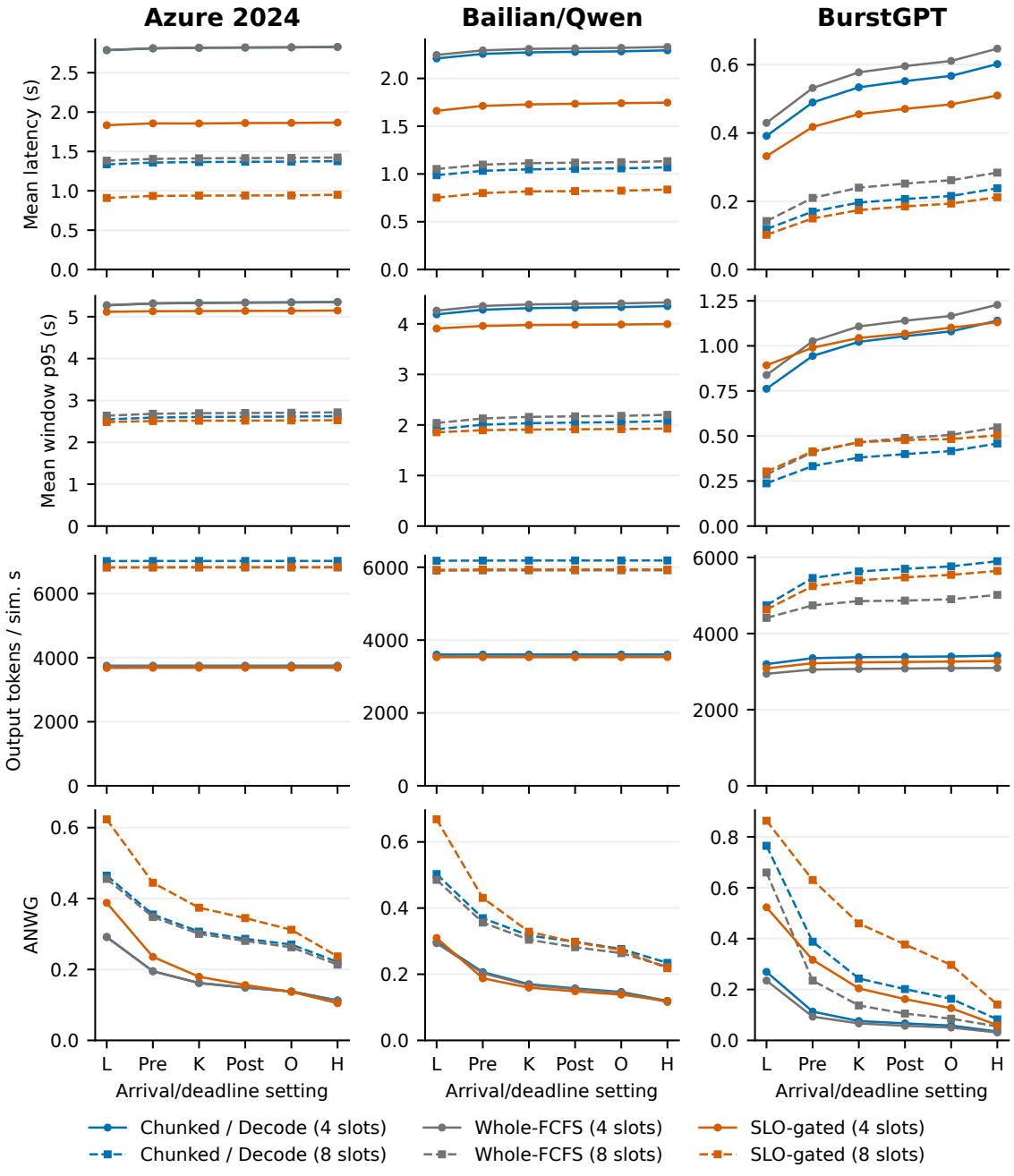


Fig. 2. Absolute performance across all six inherited settings, equally averaged over 40 windows per source. Solid circles: four slots; dashed squares: eight slots. L/Pre/K/Post/O/H abbreviate LOW through HIGH_PRESSURE in their declared order. Chunked and decode-first curves coincide exactly. The p95 panels average window-level p95s. Both arrivals and synthetic deadlines change along the horizontal axis. Lines connect descriptive condition means, with no inferential uncertainty bands.

5.4 A saved-record explanation of the trade-off

Consider BurstGPT window `burstgpt_pilot_v2_w12`, setting `HIGH_PRESSURE`, capacity 262,144 (eight resident slots). We selected this illustration retrospectively: among BurstGPT conditions where SLO-gated has lower mean latency, lower throughput and higher p95 than chunked-FCFS, it is first in lexical window/setting/capacity order. This is an outcome-selected example, not a random or representative window. Its frozen identity and verification records accompany the reproduction evidence (Appendix B).

Both adaptations complete the same 200 requests and emit 3,520 output tokens. Table 5 uses all requests: mean latency falls by 40.60%, but p95 rises by 33.17% and throughput falls by 14.63%. The throughput denominator runs from first arrival to final completion, including drain; it grows from 1.150 to 1.347 s. The p95 linearly interpolates ordered per-request latencies at zero-based position 189.05. It is a single-condition quantile, not the source mean of window p95s in Table 4.

Table 5. One outcome-selected BurstGPT condition, with identical completed inputs and output count. All times and rates are CM quantities. Duration includes the drain; percentages in the text are relative to chunked-FCFS.

Adaptation	Mean (ms)	p95 (ms)	Duration (s)	Tokens/s
Chunked-FCFS	61.681	110.741	1.150	3060.870
SLO-gated	36.641	147.480	1.347	2613.215

The saved journals show how admission competition remains active despite the nonbinding token budget. At 14 ms, request 5 has just completed, leaving one slot; requests 13 and 14 have both arrived. Chunked-FCFS admits request 13 (214 prompt tokens), whereas SLO-gated admits request 14 (201 tokens). The other request enters at 16 ms in each run. This is the first difference in admitted membership, not merely in the order of spans within a batch. Request 26 has a 1,976-token prompt and 1,120 output tokens. Its first prefill starts at 26 ms under chunked-FCFS and 223 ms under SLO-gated; its first output occurs at 31 and 228 ms, respectively. It completes last in both runs, at 1.150 and 1.347 s. Thus its later service accounts exactly for the longer denominator in this condition, without changing total output. The admission rule observes its prompt length, not its output demand. Across all requests, 151 complete earlier under SLO-gated, 29 later and 20 at the same time; the sum of latency changes is −5.008 s. Mean latency therefore improves even while the last completion and upper quantile worsen. The p95 brackets involve other requests, so the final request alone does not explain the p95 change. This illustration explains compatibility of the objectives; ABL01 supplies the controlled intervention across the corpus. The full request timeline remains in the supplementary evidence.

The example manifest records exact input/result/journal identities and SHA-256 hashes. Independent event reconstruction checks all 400 terminal states and 2,497 events against the displayed tables; the complete records support reproduction of the illustration without selecting a new condition.

5.5 Capacity and inherited-setting effects

Increasing reservation capacity doubles permitted residency from four to eight. In every source–adaptation aggregate shown, mean and mean-p95 latency fall and throughput rises. For chunked-FCFS, Azure mean latency falls from 2.8132 to 1.3626 s while throughput rises from 3,746.2 to 7,018.7 tokens/s; BurstGPT changes from 0.5225 to 0.1907 s and from 3,354.9 to 5,532.1 tokens/s. The capacity axis is predominantly a concurrency constraint in this model. The largest observed used KV is 47,872 tokens in the four-slot profile and 62,880 in the eight-slot profile, both below even the

smaller nominal capacity. These results do not measure native allocator pressure, eviction, or the benefit of paging at an exhausted used-token limit.

Figure 2 separates settings within each source–capacity. The inherited setting labels can produce large changes in goodput with small changes in latency or throughput. For Azure chunked-FCFS at four slots, LOW to HIGH_PRESSURE changes mean latency from 2.7858 to 2.8269 s and throughput from 3,745.9 to 3,746.3 tokens/s, while ANWG falls from 0.291375 to 0.112625. Since the transformation also contracts deadline slack, this is not evidence for an isolated arrival-load effect on goodput.

5.6 Matched-window variation

Figure 3 displays all 40 window-level matched differences per source and capacity, averaging the six paired setting differences within each window. It complements the source means by showing where a performance difference changes sign or is concentrated. The vertical zero line represents equal performance; it is neither a confidence bound nor the 10% practical-reversal threshold. For latency, a negative difference favors SLO-gated; for throughput, it favors chunked-FCFS.

At the unaveraged condition level, SLO-gated uniquely minimizes mean latency in all 480 Azure conditions, 468 of 480 Bailian conditions, and only 198 of 480 BurstGPT conditions. Its favorable source aggregates therefore do not mean that it wins every window/setting comparison. The tail and throughput distributions make the same distinction visible without reducing each comparison to a winner label.

Performance component of RQ3. The surviving behaviors exchange average response time against throughput, tail latency, and deadline attainment. The capacity contrast is principally a residency constraint; the setting contrast jointly stresses arrivals and deadlines. These are trade-offs within the declared model.

5.7 A prospective token-budget sensitivity

Before the reachable-state proof, we prospectively predicted that a smaller batch budget would separate chunked-FCFS from decode-first by allowing prefills to displace resident decodes. The unchanged engine requires every whole prompt to fit the budget. We therefore selected the first three BurstGPT windows in lexical order with maximum prompts above 512 and at most 1,024 tokens: `w16`, `w23`, and `w28`. All 200 requests, seeds, timing, and reservation rules remain fixed. The 72 cells combine these windows, LOW/HIGH_PRESSURE, eight slots, budgets 32,768/2,048/1,024, and four adaptations; the 12 pilot cells are reused. Every cell passes the event referee and completes all requests; the 24 baseline cells reproduce their archived outcomes.

The prediction is rejected: all 12 perturbed chunked/decode-first matches retain identical normalized journals and terminal states. Theorem 3.2 now explains why truncation can affect only the youngest resident prefill. The perturbation nevertheless changes execution. At budget 1,024, SLO-gated omits 42 of 35,976 ready-decode opportunities, with maximum anchor age 2 ms. Equation 4 gives respective 38, 51, and 44 ms upper bounds, below the 495 ms gate. Independent reconstruction finds no activation in any of its 18 cells. No strict 10% reversal occurs in 576 eligible budget comparisons, and all throughput changes are below 0.4% in magnitude. Complete effects and diagnostics remain in the supplement. These selected-window observations check conformance; the fixed-duration theorem supplies the broader equivalence result only under its stated assumptions.

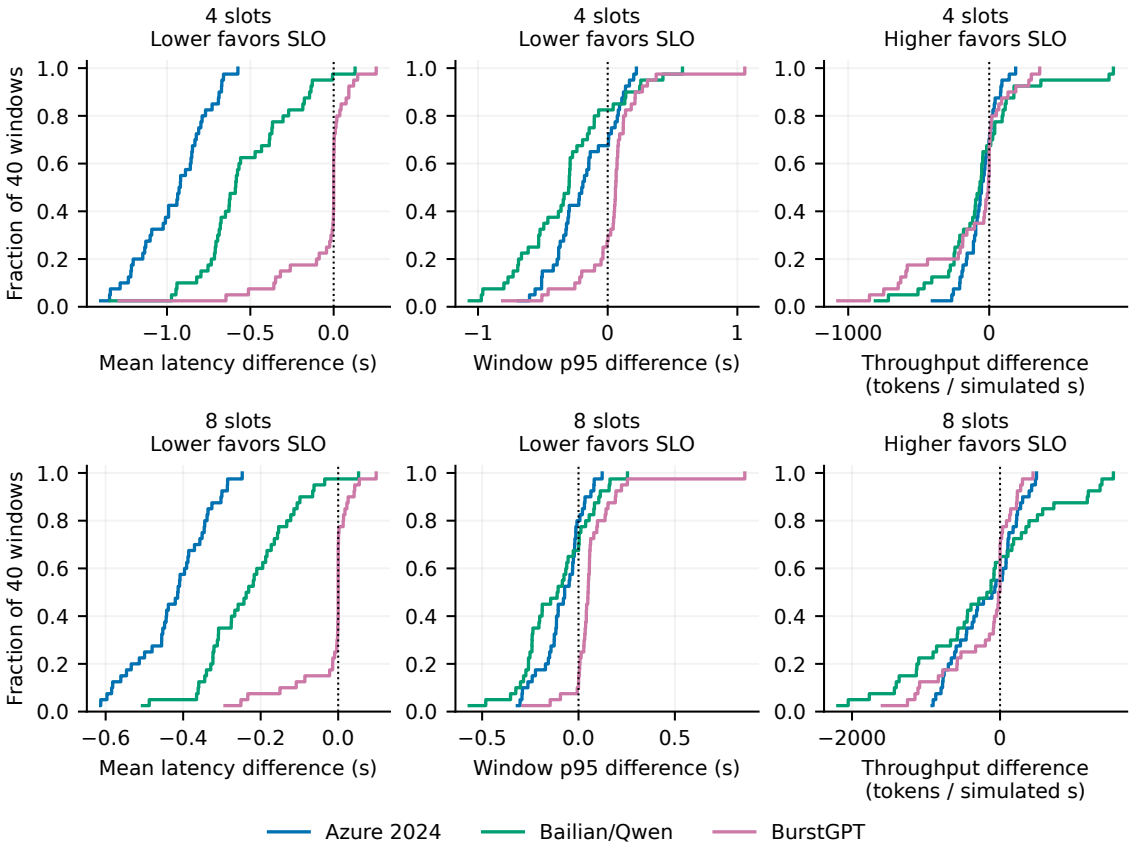


Fig. 3. Empirical distributions of SLO-gated minus chunked-FCFS differences. Each curve has 40 equally weighted windows; each point averages six matched setting differences within one window. The axes show absolute differences, not percentages. Positive latency differences and negative throughput differences are unfavorable to SLO-gated. All windows are retained.

5.8 Composition-dependent timing sensitivity

TIME01 changes batch progression to

$$T(P_b, D_b) = \left(0.5 + 0.5\frac{P_b}{512} + 0.5\frac{D_b}{8}\right) \text{ms}, \tag{8}$$

where P_b and D_b are executed prefill and decode tokens after resource constraints. Arrivals within a batch become eligible at its actual end; completion and anchor updates use that end. Equation 8 is a synthetic timing function, not hardware-calibrated. The study pairs 384 variable-duration executions with existing fixed-duration results: eight deterministically selected windows per source, two settings (LOW/HIGH_PRESSURE), two capacities, and four adaptations. Recursive stratification by mean prompt length, mean output length, and simultaneous arrival fraction selects a median window identifier per bin. This is purposive descriptor coverage, not a probability sample or evidence of blinding to prior outcomes; settings are not recalibrated for equal utilization.

Table 6. Timing sensitivity: percentage changes in chunked-FCFS means from fixed to variable timing (left), and SLO-gated relative to chunked-FCFS under variable timing (right). Means equally weight eight windows and two settings per source at each capacity. The p95 column averages window latency p95s. Negative latency and positive token-rate changes are favorable; entries are ratios of means.

		Timing change (%)		SLO vs. chunked, variable (%)		
Source	Slots	Mean	Token rate	Mean	p95	Token rate
Azure	4	-20.35	+26.82	-35.70	-3.64	-0.92
Azure	8	+10.57	-4.17	-33.47	-3.81	-1.61
Bailian	4	-22.38	+32.27	-28.27	-9.88	-2.18
Bailian	8	+7.86	+4.55	-24.95	-4.70	-3.93
BurstGPT	4	-27.24	+28.21	+2.12	+9.68	-2.43
BurstGPT	8	+1.85	+4.71	+1.88	+12.73	-2.90

All 96 chunked-FCFS/decode-first pairs retain exactly equal normalized event trajectories, timestamps, and terminal metrics. In this nonbinding-budget design, common selected work has the same order-invariant duration and hence the same successor state. This explains the observed coupling without extending the fixed-duration Theorem 3.2. Table 6 separates timing effects from policy contrasts. Using the strict loser-relative > 10% rule of Section 4.4, there are no timing reversals in 792 eligible source-mean comparisons. At window level, two of 6,309 eligible comparisons reverse (27 zero-denominator exclusions). Both compare whole-FCFS with SLO-gated in BurstGPT window `w30`, LOW, eight slots: p95 latency and p95 TTFT. They are correlated tail endpoints in one condition, not independent replications. Absence of strict reversals does not imply identical rankings.

On the selected BurstGPT windows, SLO-gated mean latency is 2.12% and 1.88% higher at four/eight slots under variable timing; it is already 2.41% and 2.29% higher under fixed timing. This is subset sensitivity, not a timing-induced policy reversal. SLO-gated urgency remains inactive; the timing study establishes neither hardware benefits nor performance outside the selected windows.

6 Rankings, Aggregation, and Metric Choice (RQ3)

6.1 Stable practical source comparisons and heterogeneous windows

Table 7 reports the strict-10% reversal rule with eligible and total denominators. Across all source-mean comparisons the count is **zero out of 2,376**. Each source mean uses all 40 windows at a fixed setting, capacity, and metric. The denominator is 3 source pairs ×6 settings ×2 capacities ×11 endpoints ×6 policy pairs. For this equal-window estimand, changing source does not overturn a policy-pair preference by more than the declared margin in both directions. The 2,376 comparisons include the equivalent chunked/decode-first pair and redundant throughput endpoints; they do not constitute 2,376 independent confirmations or establish population equivalence.

This null result is compatible with smaller sign changes and different tie sets. Among 396 comparisons of source pairs at fixed settings, capacities, and metrics, exact top sets are equal in 307 and intersect in 323; complete average-rank vectors are equal in 247. Absence of a strict practical reversal is weaker than equality of rankings or top sets.

Window-level variation remains visible. Within-source window comparisons yield 34,394 reversals among 1,847,700 eligible comparisons; Cartesian cross-source window comparisons yield 109,913 among 3,788,640. For ANWG alone, the corresponding counts are 8,472/162,900 and 35,352/332,640. The larger window-pair denominators arise from comparing every unordered pair of 40 windows within a source (780) or every Cartesian pair across sources (1,600). They are not

Table 7. Strict-10% practical reversals. Nonzero ties remain in eligible denominators; a zero losing value is unestimable. All counts include six policy pairs and the declared endpoints. Reused windows and overlapping pairs make these dependent descriptive counts, not independent trials.

Comparison axis	Reversals	Eligible	Total	Unestimable
Within-source windows	34,394	1,847,700	1,853,280	5,580
Cross-source windows	109,913	3,788,640	3,801,600	12,960
Inherited settings	920	236,853	237,600	747
KV capacities	350	47,418	47,520	102
Metric pairs	11,795	473,580	475,200	1,620
Source means	0	2,376	2,376	0

Table 8. Ranking summaries, each out of 1,440 window/setting/capacity conditions. "Near-tie change" counts conditions whose top set changes under the predeclared $10^{-12}\max(1, |v_{\text{best}}|)$ tolerance. No ranking is missing. "SLO unique top" always uses exact equality.

Endpoint	Exact top tie	SLO unique top	Near-tie change
ANWG	425	1,004	0
Mean latency	289	1,146	6
p95 latency	586	834	0
Token throughput	906	482	0
Completion	1,440	0	0
Mean TPOT	979	461	1,129
p95 TPOT	1,348	92	104

larger independent sample sizes. These summaries cannot establish that one axis causally contributes more instability than another.

Within the same window, settings produce 920/236,853 eligible reversals, capacity changes 350/47,418, and metric choice 11,795/473,580. The last count compares policy-pair directions across endpoints; it demonstrates that a preference can depend on the objective even with the workload and resources fixed. It is distinct from comparing top sets, which asks whether at least one best policy is shared rather than whether a particular pair reverses.

6.2 Ties and redundant endpoints

Table 8 retains exact ties instead of breaking them by policy identifier. SLO-gated is uniquely best for mean latency in 1,146 conditions and for throughput in 482. Completion has a four-way tie everywhere.

All requests finish, making emitted-token and completed-token throughput identical. Request throughput differs only by a workload-specific multiplier shared by all policies at a fixed condition. The three throughput rankings therefore coincide. Chunked-FCFS, decode-first, and SLO-gated also produce essentially one output per millisecond after decoding begins: the largest chunked-versus-SLO mean-TPOT difference is 1.80×10^{-16} s. Timestamp subtraction nevertheless changes exact top sets; the predeclared near-tie tolerance changes mean-TPOT top sets in 1,129 conditions and p95-TPOT top sets in 104. These tiny distinctions have no practical performance meaning.

Across all 79,200 metric-pair ranking comparisons, exact top sets are equal in 27,915 and intersect in 57,602. These counts include redundant throughput endpoints, completion ties, and numerical TPOT distinctions. They summarize reporting consequences; they are not eleven independent tests of a scheduler's robustness.

Table 9. Post hoc reduced view: three behavior pairs and eight endpoints. The original strict-10% rule is unchanged. These dependent descriptive counts supplement, rather than replace, Table 7.

Axis	Reversals	Eligible	Total
Source means	0	864	864
Within-source windows	18,732	670,637	673,920
Cross-source windows	61,880	1,374,720	1,382,400
Settings	629	85,962	86,400
Capacity	240	17,219	17,280
Metric	4,492	120,288	120,960

6.3 Post hoc view of distinct behaviors and metric families

To test whether duplication drives the qualitative ranking conclusions, we project the saved outputs onto whole-FCFS, chunked-FCFS, and SLO-gated. We remove the universally tied completion endpoint and retain token throughput as the representative of the three identical throughput rankings. The resulting eight endpoints cover five families: latency, TTFT, TPOT, throughput, and deadline attainment. Both mean and p95 remain for the three time families because they describe different distributional features. TPOT remains informative for whole-FCFS: its mean and p95 exceed the best value by the same strict-10% practical margin in 341 and 1,027 conditions, respectively. Removing TPOT altogether would discard that contrast.

This view contains 11,520 rankings, 34,560 matched pairwise records, and 8,208 reversal groups. Independent exact-boundary calculations from absolute metrics reproduce every retained archived group. Table 9 still has no source-mean practical reversals (0/864: three source pairs, six settings, two capacities, eight endpoints, and three behavior pairs), but has window and metric reversals. This is stability of equal-window source means under the strict-10% rule, not universal ranking portability. The qualitative contrast therefore survives removal of these demonstrated redundancies. Rates change with the denominator; the remaining endpoints and overlapping comparisons are not independent replications. The original four-label, eleven-endpoint outputs remain intact. Chronological sensitivities appear in Appendix C.

Ranking component of RQ3. Source-mean practical stability does not imply uniform preferences across windows or objectives. This remains true after collapsing duplicate labels and endpoints; magnitudes and top sets distinguish meaningful preferences from ties and numerical noise.

7 Supporting Historical Interpretation and Calibration

7.1 What the historical comparison establishes

The historical analysis reported 36 of 990 policy-pair/source-pair/setting records meeting its practical-margin and nominal-interval rule. Numerical reproduction confirms that archived-rule count. It does not establish population reversals: the historical policy–model combinations differ, selected trace windows have no known population sampling law, and the inference procedure's calibration was unestablished. The C01 panel and execution model change the comparison object, so its zero source-mean practical reversals are not a like-for-like disappearance of those 36 records. Historical and common-model outcomes are kept in separate evidence strata rather than combined into a leaderboard.

Table 10. Compact calibration diagnostic for the actual historical-source screened-BH comparator. Each row has 8,000 studies; ordinary fixed-contrast interval coverage is available in all of them. Bounds are the expanded 17-method simultaneous Monte Carlo bounds, not confidence intervals for C01 performance. These three null scenarios isolate IID, independent-block, and autoregressive (AR) dependence; all 18 scenarios and methods remain in the supplement.

Scenario	FDR	MC bounds	Coverage	MC bounds
IID equal null	0.09025	[0.06878, 0.11172]	0.94187	[0.92357, 0.96018]
Independent-block null	0.91800	[0.89726, 0.93874]	0.64762	[0.61465, 0.68060]
AR(0.7) null	0.98037	[0.96780, 0.99295]	0.56487	[0.53081, 0.59894]

7.2 A focused known-truth diagnostic

STAT01 evaluates historical inference in synthetic studies whose population means, covariance, and missingness are specified. Each study has three sources, eleven policies, and normally 40 windows per source; one bounded scenario uses 400. The 165 source-pair/policy-pair hypotheses are dependent within a study. Independently generated studies, not their individual hypotheses, are the Monte Carlo units. The completed design contains 134,000 studies across 18 scenarios, including independent and identically distributed (IID), block-dependent, autoregressive, nonnormal, bounded, and missing-data cases.

The primary truth is a population reversal exceeding 10% relative to the losing mean in both sources. Sign reversals are a separate endpoint. The historical procedure first screens noisy point estimates at 10%, then uses bootstrap intervals and zero-effect p-values on the selected contrasts. Testing a zero-effect null after such screening is not automatically a test of the practical-effect null. Benjamini–Hochberg (BH) or Benjamini–Yekutieli (BY) adjustment alone cannot supply the missing calibration [8, 9].

Three implementation identities are retained. The eight frozen original procedures include a port of the historical screened-BH procedure. A post-diagnosis comparator reproduces the actual historical source's floating-point evaluation order. Eight additional comparisons implement explicit mathematical threshold semantics. Thus the expanded family has 17 methods, with nine additions identified as post-diagnosis. Arithmetic correction is not a new test and does not change the inferential assumptions. Appendix A gives the definitions and the saved boundary case.

Table 10 shows that the actual historical-source procedure has practical false-discovery rate (FDR) 0.09025 in the IID equal-null scenario, with an expanded Monte Carlo lower bound above 0.05. Independent-block and AR nulls show much larger false-discovery rates and poor ordinary interval coverage. These null examples already demonstrate a calibration concern without requiring interpretation of power or near-boundary alternatives.

Appendix A retains all classifications, stopping counts, original-eight and expanded-17 uncertainty families, abstentions, precision gaps, and low-power outcomes. These qualify the calibration findings.

Implication for the common-contract study. Reproducing historical arithmetic establishes what its rule reported; execution conformance establishes that C01 follows its common model. Neither establishes valid statistical inference. STAT01 demonstrates calibration failures for historical procedures in explicit known-truth scenarios, while retaining uncertainty and assumption-specific limits. C01's finite-corpus descriptions do not depend on population significance tests.

8 Discussion and Threats to Validity

From policy labels to interpretable comparisons. The structural analysis identifies which decision rules can change execution before interpreting their rankings. The resident-prefix invariant and common batch-end updates erase the chunked/decode-first priority distinction, even under binding budgets; admission competition and whole-prefill/decode exclusion survive. The urgency certificate supplies a different kind of result: a bound on reachable anchor age rules out an otherwise implemented branch. Together these explain why a nominally richer policy panel need not represent more executed mechanisms.

ABL01 identifies prompt ordering as the source of the residual latency–throughput–deadline trade-off within the nonbinding contract. This is a total intervention effect, not a decomposition of admission, queueing, and batch-count mediators or a hardware-level causal claim. The saved BurstGPT timeline illustrates how many earlier completions can lower mean latency while a later final completion extends the drain. It does not establish optimality or a corpus-wide causal decomposition. Source means answer an equal-window average question; window reversals expose variation concealed by that average, while cross-metric reversals concern different objectives on the same workload. Their coexistence, even in the reduced analysis, shows why stable aggregate preferences cannot select a scheduler for every window and objective. Exact top sets further distinguish ties from strict preferences.

Model adequacy and transfer. Fixed batch timing and conservative full-context reservations permit the state-transition argument but omit composition-dependent efficiency, kernel, bandwidth, communication, and native allocation costs. Theorem 3.2 permits binding budgets but depends on fixed policies, an empty initial system, resident-first service, stable prefix admission, and the specified progress rules. It does not establish invariance under policy switching or arbitrary supplied states. The budget sensitivity changes some admissions and SLO-gated decode service; the finite-work certificate rules out urgency activation there. Longer prompts fail the unchanged domain check at these smaller budgets. TIME01 tests one synthetic composition-dependent timing function on 24 selected windows. It neither calibrates hardware costs nor establishes robustness to arbitrary timing functions. ABL01's control coupling is restricted to the original nonbinding budget and admission semantics; it does not automatically extend to binding-budget SLO-gated execution. Historical GPU studies do not validate M3's exact comparisons. Conformance checks establish implementation correctness for the abstraction, not physical prediction accuracy [21].

Corpus and inference limits. The selected windows are finite and temporally related, and the Azure-linked sources are not independent provider populations. Nonoverlap inside the C01 panel does not settle overlap with older cross-project BurstGPT selections. Uniform priorities and synthetic output-derived deadlines restrict goodput interpretation; the inherited transformation changes deadlines and arrivals together. C01 consequently reports descriptive finite-corpus effects. The timing subset makes this limit concrete: on all forty BurstGPT windows at the same two settings, fixed-duration SLO-gated improves mean latency by 15.24% and 11.99% at four/eight slots. The eight selected windows instead worsen it under both clocks (Section 5.8). Input composition differences are descriptive, not an identified explanation; neither subset nor full-corpus averages estimate an unmeasured population. The reduced analysis is post hoc and removes only demonstrated redundancies; its remaining metric families and overlapping comparisons are still dependent.

STAT01's known truth applies only to its generators. Post-diagnosis methods, finite study counts, conditional interval availability, and conservative power remain explicit in Appendix A. Reproducible arithmetic and valid population

inference are distinct requirements. The extensive checks reduce implementation and reporting risk without adding scientific novelty or resolving model adequacy by themselves.

9 Conclusion

An admission-and-progress invariant proves selected-work equivalence of fixed chunked-FCFS and decode-first adaptations from an empty serving system under the declared contract, including binding budgets. Common batch-end updates extend this equality to normalized transitions and terminal outcomes. A separate finite-work certificate explains urgency inactivity in the evaluated fixed-duration regimes. The controlled ablation attributes the remaining trade-offs to prompt ordering within the original contract. Synthetic timing sensitivity preserves observed pair equivalence but reveals window-sensitive policy effects. Source-mean practical stability coexists with local and cross-metric reversals after demonstrated redundancies are removed. These results connect model constraints, effective behavior, and the interpretation of performance comparisons; they establish neither scheduler optimality nor hardware performance.

Code and Data Availability

Code and reproduction scripts for `m3-v1.0.0` are available in the GitHub release. Result tables are distributed on Hugging Face, with an identical-data mirror on Kaggle. The complete saved-evidence archive, including provenance and component-specific licenses, is preserved in Zenodo (doi:10.5281/zenodo.23608723). Reproducing analyses of saved evidence is distinct from rerunning the full campaign, which requires upstream inputs and the documented environment. Original third-party raw traces are excluded and must be obtained from the attributed providers. Original software is MIT-licensed; eligible original results and dataset documentation use CC BY 4.0. Upstream terms and author copyright in the manuscript are retained.

Declarations

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AI writing assistance. OpenAI Codex assisted with manuscript revisions. OpenAI ChatGPT assisted with manuscript organization and suggestions for presenting and explaining the research. Research-related assistance and the author's review and validation workflow are described in Section 3.6. The author reviewed the resulting work and accepts full responsibility for the manuscript's content, scientific accuracy, and conclusions.

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References

[1] Amey Agrawal, Anmol Agarwal, Nitin Kedia, Jayashree Mohan, Souvik Kundu, Nipun Kwatra, Ramachandran Ramjee, and Alexey Tumanov. 2024. Etalon: Holistic Performance Evaluation Framework for LLM Inference Systems. arXiv:2407.07000v2. https://arxiv.org/abs/2407.07000v2

[2] Amey Agrawal, Nitin Kedia, Anmol Agarwal, Jayashree Mohan, Nipun Kwatra, Souvik Kundu, Ramachandran Ramjee, and Alexey Tumanov. 2025. On Evaluating Performance of LLM Inference Serving Systems. arXiv:2507.09019v1. doi:10.48550/arXiv.2507.09019

[3] Amey Agrawal, Nitin Kedia, Jayashree Mohan, Ashish Panwar, Nipun Kwatra, Bhargav S. Gulavani, Ramachandran Ramjee, and Alexey Tumanov. 2024. VIDUR: A Large-Scale Simulation Framework for LLM Inference. In *Proceedings of Machine Learning and Systems (MLSys '24)*, Vol. 6. MLSys, 351–366. https://proceedings.mlsys.org/paper_files/paper/2024/hash/b74a8de47d2b3c928360e0a011f48351-Abstract-Conference.html

[4] Amey Agrawal, Nitin Kedia, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani, Alexey Tumanov, and Ramachandran Ramjee. 2024. Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 117–134. https://www.usenix.org/conference/osdi24/presentation/agrawal

[5] Alibaba Cloud. 2026. *qwen-bailian-usagetraces-anon.* https://github.com/alibaba-edu/qwen-bailian-usagetraces-anon/tree/5f7439c51ec248a0c585f7d90a41a6f57773b912 Anonymized production usage-trace release at the pinned URL; Apache License 2.0; consumed trace identity verified against the pinned Git LFS object.

[6] Ruicheng Ao, Gan Luo, David Simchi-Levi, and Xinshang Wang. 2026. Optimizing LLM Inference: Fluid-Guided Online Scheduling with Memory Constraints. arXiv:2504.11320v4. doi:10.48550/arXiv.2504.11320 First submitted April 15, 2025; version 4, June 13, 2026.

[7] Agrim Bari, Parikshit Hegde, and Gustavo de Veciana. 2025. Optimal Scheduling Algorithms for LLM Inference: Theory and Practice. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 9, 3, Article 59 (2025), 43 pages. doi:10.1145/3771574

[8] Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 1 (1995), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x

[9] Yoav Benjamini and Daniel Yekutieli. 2001. The Control of the False Discovery Rate in Multiple Testing under Dependency. *The Annals of Statistics* 29, 4 (2001), 1165–1188. doi:10.1214/aos/1013699998

[10] Jaehong Cho, Hyunmin Choi, Guseul Heo, and Jongse Park. 2026. LLMServingSim 2.0: A Unified Simulator for Heterogeneous and Disaggregated LLM Serving Infrastructure. In *2026 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*. IEEE, Seoul, South Korea, 1–14. doi:10.1109/ISPASS69572.2026.00012

[11] Jaehong Cho, Minsu Kim, Hyunmin Choi, Guseul Heo, and Jongse Park. 2024. LLMServingSim: A HW/SW Co-Simulation Infrastructure for LLM Inference Serving at Scale. In *2024 IEEE International Symposium on Workload Characterization (IISWC)*. IEEE, Vancouver, BC, Canada, 15–29. doi:10.1109/IISWC63097.2024.00012

[12] J. G. Dai, Tianze Deng, Yueying Li, and Tianyi Peng. 2026. Throughput-Optimal Scheduling Algorithms for LLM Inference and AI Agents. arXiv:2504.07347v3. https://arxiv.org/abs/2504.07347v3 Version 3, May 18, 2026.

[13] Yichao Fu, Siqi Zhu, Runlong Su, Aurick Qiao, Ion Stoica, and Hao Zhang. 2024. Efficient LLM Scheduling by Learning to Rank. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Vancouver, BC, Canada, 59006–59029. doi:10.52202/079017-1882

[14] Wassily Hoeffding. 1963. Probability Inequalities for Sums of Bounded Random Variables. *J. Amer. Statist. Assoc.* 58, 301 (1963), 13–30. doi:10.1080/01621459.1963.10500830

[15] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. https://www.jstor.org/stable/4615733

[16] Ke Hong, Xiuhong Li, Lufang Chen, Qiuli Mao, Xuefei Ning, Guohao Dai, Shengen Yan, Yun Liang, and Yu Wang. 2025. SOLA: Optimizing SLO Attainment for Large Language Model Serving with State-Aware Scheduling. In *Proceedings of Machine Learning and Systems*, Vol. 7. MLSys, Santa Clara, CA, 13 pages. https://proceedings.mlsys.org/paper_files/paper/2025/hash/bc82dbfbfa43232be85b8d9838f49c3e-Abstract-Conference.html

[17] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23)*. ACM, Koblenz, Germany, 611–626. doi:10.1145/3600006.3613165

[18] Andreas Maurer and Massimiliano Pontil. 2009. Empirical Bernstein Bounds and Sample Variance Penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*. 9 pages. https://www.cs.mcgill.ca/~colt2009/papers/012.pdf

[19] Microsoft Azure. 2024. *AzureLLMInferenceDataset2024.* https://github.com/Azure/AzurePublicDataset/blob/master/AzureLLMInferenceDataset2024.md

[20] Konstantinos Papaioannou and Thaleia Dimitra Doudali. 2024. The Importance of Workload Choice in Evaluating LLM Inference Systems. In *Proceedings of the 4th Workshop on Machine Learning and Systems (EuroMLSys '24)*. ACM, Athens, Greece, 39–46. doi:10.1145/3642970.3655823

[21] Robert G. Sargent. 2013. Verification and validation of simulation models. *Journal of Simulation* 7, 1 (2013), 12–24. doi:10.1057/jos.2012.20

[22] Jianshu She, Zonghang Li, Hongchao Du, Shangyu Wu, Wenhao Zheng, Eric P. Xing, Zhengzhong Liu, Huaxiu Yao, Jason Xue, and Qirong Ho. 2026. PLA-Serve: A Prefill-Length-Aware LLM Serving System. In *Proceedings of Machine Learning and Systems*, Vol. 8. MLSys, Bellevue, WA, 17 pages. https://proceedings.mlsys.org/paper_files/paper/2026/hash/bbb7506579431a85861a05fff048d3e1-Abstract-Conference.html Proceedings title; the linked paper is titled LAPS: A Length-Aware-Prefill LLM Serving System.

[23] Jovan Stojkovic, Chaojie Zhang, Íñigo Goiri, Josep Torrellas, and Esha Choukse. 2025. DynamoLLM: Designing LLM Inference Clusters for Performance and Energy Efficiency. In *2025 IEEE International Symposium on High Performance Computer Architecture (HPCA)*. IEEE, Las Vegas, NV, 1348–1362. doi:10.1109/HPCA61900.2025.00102

[24] Biao Sun, Ziming Huang, Hanyu Zhao, Wencong Xiao, Xinyi Zhang, Yong Li, and Wei Lin. 2024. Llumnix: Dynamic Scheduling for Large Language Model Serving. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 173–191. https://www.usenix.org/conference/osdi24/presentation/sun-biao

[25] Jiahao Wang, Jinbo Han, Xingda Wei, Sijie Shen, Dingyan Zhang, Chenguang Fang, Rong Chen, Wenyuan Yu, and Haibo Chen. 2025. KVCache Cache in the Wild: Characterizing and Optimizing KVCache Cache at a Large Cloud Provider. In *2025 USENIX Annual Technical Conference (USENIX ATC 25)*. USENIX Association, Boston, MA, 465–482. https://www.usenix.org/conference/atc25/presentation/wang-jiahao

[26] Yuxin Wang, Yuhan Chen, Zeyu Li, Xueze Kang, Yuchu Fang, Yeju Zhou, Yang Zheng, Zhenheng Tang, Xin He, Rui Guo, Xin Wang, Qiang Wang, Amelie Chi Zhou, and Xiaowen Chu. 2025. BurstGPT: A Real-World Workload Dataset to Optimize LLM Serving Systems. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25), Volume 2*. ACM, Toronto, ON, Canada, 5831–5841. doi:10.1145/3711896.3737413

[27] Bingyang Wu, Yinmin Zhong, Zili Zhang, Shengyu Liu, Fangyue Liu, Yuanhang Sun, Gang Huang, Xuanzhe Liu, and Xin Jin. 2026. FastServe: Iteration-Level Preemptive Scheduling for Large Language Model Inference. In *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*. USENIX Association, Renton, WA, 57–74. https://www.usenix.org/conference/nsdi26/presentation/wu-bingyang

[28] Yuxing Xiang, Xue Li, Kun Qian, Yan Zhang, Wenyuan Yu, Ennan Zhai, Xin Jin, and Jingren Zhou. 2026. ServeGen: Workload Characterization and Generation of Large Language Model Serving in Production. In *Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)*. USENIX Association, Renton, WA, 1845–1859. https://www.usenix.org/conference/nsdi26/presentation/xiang-servegen

[29] Gyeong-In Yu, Joo Seong Jeong, Geon-Woo Kim, Soojeong Kim, and Byung-Gon Chun. 2022. Orca: A Distributed Serving System for Transformer-Based Generative Models. In *Proceedings of the 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI '22)*. USENIX Association, Carlsbad, CA, 521–538. https://www.usenix.org/conference/osdi22/presentation/yu

[30] Netanel Zakay and Dror G. Feitelson. 2014. Workload resampling for performance evaluation of parallel job schedulers. *Concurrency and Computation: Practice and Experience* 26, 12 (2014), 2079–2105. doi:10.1002/cpe.3240

[31] Wei Zhang, Zhiyu Wu, Yi Mu, Rui Ning, Banruo Liu, Nikhil Sarda, Myungjin Lee, and Fan Lai. 2026. JITServe: SLO-aware LLM Serving with Imprecise Request Information. In *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*. USENIX Association, Renton, WA, 825–848. https://www.usenix.org/conference/nsdi26/presentation/zhang-wei

[32] Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, and Hao Zhang. 2024. DistServe: Disaggregating Prefill and Decoding for Goodput-Optimized Large Language Model Serving. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 193–210. https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin

A Statistical Diagnostic: Definitions and Complete Scenario Summary

A.1 Study construction and truth

A study draws a latent array $Y \in \mathbb{R}^{3\times n\times 11}$, with positive population means μ_{sj}. Sources are independent within this synthetic design. Dependence across policies is $C(\rho) = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^{\mathsf{T}}$, usually with $\rho = 0.5$. Temporal covariance is identity, independent five-window blocks with within-block correlation 0.7, or AR(1) with correlation $0.7^{|i-j|}$. Source-specific innovations have mean zero and variance one; a Cholesky transformation imposes the specified covariance, with default scale $\sigma = 0.15$. The exact-ties scenario uses $\rho = 1$.

Each family comprises $\binom{3}{2}\binom{11}{2} = 165$ hypotheses. For a source pair (s,t) and policy pair (a,b), sign-reversal truth is $(\mu_{sa} - \mu_{sb})(\mu_{ta} - \mu_{tb}) < 0$. Practical-reversal truth is given by Equation 9:

$$\begin{aligned}(\mu_{sa} > 1.1\mu_{sb} \ \wedge \ \mu_{tb} > 1.1\mu_{ta}) \\ \vee \ (\mu_{sb} > 1.1\mu_{sa} \ \wedge \ \mu_{ta} > 1.1\mu_{tb}).\end{aligned} \tag{9}$$

Equality at 10% is null. Truth is enumerated from decimal/rational mean specifications, independently of estimated means and test decisions.

Equal-null means are 0.5. Stable and near-tie means are $0.5 + 0.01j$ and $0.5 + 0.0002j$, respectively. A reversal of size δ assigns source 0 policies 0/1 means $0.5 \pm \delta/2$, swaps them in source 1, and sets all other means to 0.5. Small, medium, and large IID reversal scenarios use $\delta = 0.03, 0.10, 0.30$. The boundary scenario uses 0.55/0.50 and the reverse. Block, AR, unequal-variance, mixed-distribution, and missing completely at random (MCAR) reversal scenarios use $\delta = 0.15$. Unequal scales are $(0.08, 0.25, 0.15)$; mixed innovations are standardized normal, Student-t_5, and lognormal. MCAR retains each policy/window observation independently with probability 0.8. Selective missingness removes source 0/policy 0 and source 1/policy 1 observations below their respective latent means; truth remains the unconditional mean. Observed values cannot identify that target merely by ignoring missingness.

The bounded scenarios use IID standardized uniform innovations, $\sigma = 0.08$, and $\delta = 0.30$, giving scores in $[0,1]$ analytically, without clipping. All scenarios have 40 windows per source except the last, which has 400. Raw arrays and

masks are retained in the complete evidence archive. The 112,275 reused and 21,725 subsequently generated studies sum to the completed 134,000-study design; no study was removed because of a numerical disagreement between implementations.

A.2 Frozen methods and post-diagnosis additions

All methods use nominal level $\alpha = q = 0.05$. The historical screened procedure uses 2,000 bootstrap draws of matched observed per-window policy differences, percentile intervals, and $p = \min\{1, 2\min[\Pr_{\text{boot}}(\bar{D} \le 0), \Pr_{\text{boot}}(\bar{D} \ge 0)]\}$. It combines source-side p-values by their maximum, applies BH to the point-screened family, and requires both intervals to exclude zero. The full-family variant assigns untested hypotheses p-value one and applies BH over all 165. The historical plus-one BY variant transforms the component bootstrap p-value as $\min\{1, (2000p + 2)/2001\}$ before source combination and applies BY over 165. These modifications do not make selection or the original p-values valid by definition.

The analytic practical-effect methods use paired contrasts $Z = Y_a - 1.1Y_b$ and their reversed orientation. Each orientation combines the two one-sided source tests by a maximum (an intersection–union test). The union over orientations uses $\min\{1, 2\min(p_1, p_2)\}$ before Holm correction over 165 hypotheses [15]. IID t-Holm uses paired Student tests; cluster t-Holm uses independent five-window cluster averages, retaining only completely observed clusters. Fewer than two complete clusters causes abstention. Their finite-sample Gaussian justifications apply to their respective independent sampling units with complete or MCAR observations, not arbitrary temporal dependence. The covariance-oracle normal variant uses the known Gaussian covariance of each observed matched sample mean. It does not access missing latent outcomes and is not deployable without that covariance information.

Grouped-bootstrap BY resamples nonoverlapping five-window groups jointly across policies, preserving masks and pairing. Its centered one-sided component p-value is $(1 + \#\{\bar{Z}^* - \bar{Z} \ge \bar{Z}\})/2001$. A zero-observation resample causes component abstention; it is not silently excluded. The same orientation construction is followed by full-family BY. No finite-sample guarantee was prespecified for this procedure. Bounded Hoeffding–Holm applies only to the bounded IID scenarios. Since $Z \in [-1.1, 1]$, its component p-value is $\exp[-2m\max(\bar{Z}, 0)^2/2.1^2]$ [14], followed by the same orientation and Holm construction. It abstains outside that domain.

The eight original identities are retained unchanged. The ninth method, `legacy_source_screened_BH`, reproduces the actual frozen historical source's arithmetic. Eight `exact_` counterparts of the original methods use explicit rational critical thresholds and exact bootstrap-count grids. Analytic p-values are treated as the exact dyadic values represented by their finite floats, not as infinitely accurate analytic probabilities. Rational harmonic constants define BY thresholds. These additions change numerical semantics, not the sampling assumptions, screening target, or statistical validity of a component test.

A.3 Coverage, availability, power, and Monte Carlo accounting

For each independent study, let R be discoveries and V false discoveries. False-discovery proportion is $V/\max(R, 1)$; its expectation is FDR. Familywise error is $\mathbf{1}\{V > 0\}$, and power is true discoveries relative to the number of alternatives when alternatives exist. Both practical and sign truth are retained. Ordinary interval coverage uses a fixed source 0/policy 0-minus-policy 1 contrast. For historical methods, that interval is generated in a separate unconditional diagnostic stream. Coverage of actually selected intervals is a different, selection-conditional endpoint; it is not replaced by ordinary coverage.

Interval availability is reported separately. In the MCAR and selectively missing scenarios, cluster-t ordinary coverage is based on 1,676 and 202 of 8,000 studies, respectively. Historical selected-interval coverage in the IID equal null is 0.123834 over 958 studies emitting selected intervals, whereas ordinary fixed-contrast coverage is 0.941875 over 8,000 studies. These are study-level averages, not independent interval observations. The complete endpoint CSV also includes null rejection fractions, maximizer-set coverage and size, and deterministic-ID correctness; exact empirical maximizing sets are not confidence sets.

The five possible analysis looks are 500, 1,000, 2,000, 4,000, and 8,000 studies per scenario. For a bounded study-level endpoint with sample variance s^2 and available count N, the two-sided empirical Bernstein half-width is Equation 10:

$$b = \sqrt{\frac{2s^2 \log(4/\delta)}{N}} + \frac{7 \log(4/\delta)}{3(N-1)}. \tag{10}$$

The bounds clip $\bar{X} \pm b$ to $[0, 1]$; empty endpoints are unestimated with the full range. The construction uses the empirical Bernstein bound and a finite union over both tails and all reported looks [18]. Conditional coverage uses the count of IID studies emitting an interval, with that availability explicit. The original allocation is $\delta = 0.05/(18 \cdot 8 \cdot 16 \cdot 5)$; the expanded report uses $0.05/(18 \cdot 17 \cdot 16 \cdot 5)$. Dependence among methods and endpoints does not invalidate the union bound. The wider accounting does not erase post-diagnosis method selection.

Stopping uses only the original eight methods. Required half-widths are 0.025 for practical FDR, familywise error, and null rejection fraction; 0.03 for ordinary interval coverage; and 0.08 for defined power. Selected-interval coverage is not a stopping target. The calibration screen flags concern if the practical-FDR lower bound exceeds 0.05 or ordinary-coverage upper bound is below 0.95. Empirical compatibility with tolerance requires FDR upper bound at most 0.06 and coverage lower bound at least 0.94 in relevant assumption-valid scenarios. All other estimable cases are inconclusive. The screen permits conservative overcoverage and is not a proof of exact nominal calibration.

Table 11 summarizes the original-method scenario screens and stopping counts.

A.4 Diagnostic outcomes and stopping interpretation

Other scenarios expose the practical-versus-sign target mismatch: practical FDR is 0.39500 at the exact practical boundary and approximately 0.73316 in the medium-reversal scenario. Full scenario results are retained rather than summarized as a single global error rate.

These findings also have limits. Assumption-aligned Gaussian procedures frequently remain *inconclusive* under the stringent simultaneous Monte Carlo screen; that is not evidence that they are miscalibrated. Conservative bounded Hoeffding–Holm comparisons have zero observed false discoveries and 100% observed diagnostic coverage, but practical power is zero at 40 windows and approximately 0.05263 at 400. The campaign does not certify a generally useful replacement procedure. Of 144 original method/scenario classifications, 45 indicate concern, 74 are inconclusive, 9 meet the empirical tolerance screen, and 16 abstain outside their domain. Seven of the nine compatible entries arise in the degenerate exact-ties scenario; the remaining two are bounded Hoeffding.

Completion and precision must also be separated. Scenarios 3 and 17 stop at 2,000 and 4,000 studies; the other sixteen finish at 8,000. Six scenarios meet the original precision objective, while twelve stop at the maximum count with precision gaps. The original eight-method uncertainty family controls stopping. The expanded 17-method report has its own, wider allocation; additions do not retrospectively become prespecified. Neither maximum-count completion nor an inconclusive calibration screen is a successful calibration claim.

Table 11. Complete original-eight scenario screen and stopping counts. C: calibration concern; I: inconclusive; E: empirical compatibility with tolerance; A: outside-domain abstention. P: original precision objective met; M: maximum-count stop with gaps. H-S/H-F/H-BY: historical screened BH/full BH/plus-one BY; t/c-t/O/G/Hf: IID t/cluster t/oracle normal/grouped bootstrap/bounded Hoeffding. These use the *original-eight* uncertainty family. Full numerical results for all 17 methods and both families are in `stat01-all-endpoints.csv`.

ID	Scenario	Studies	Stop	H-S	H-F	H-BY	t	c-t	O	G	Hf
0	iid_equal_null	8,000	P	C	I	I	I	I	I	C	A
1	iid_stable_null	8,000	P	I	I	I	I	I	I	C	A
2	iid_near_tie_null	8,000	P	C	I	I	I	I	I	C	A
3	exact_ties_null	2,000	P	E	E	E	E	E	E	E	A
4	practical_boundary	8,000	M	C	I	I	I	I	I	C	A
5	small_sign_reversal	8,000	M	C	I	I	I	I	I	C	A
6	medium_reversal	8,000	M	C	C	I	I	I	I	C	A
7	large_reversal	8,000	M	I	I	I	I	I	I	C	A
8	block_null	8,000	M	C	C	C	C	I	I	C	A
9	block_reversal	8,000	M	C	C	C	C	I	I	C	A
10	ar_null	8,000	M	C	C	C	C	C	I	C	A
11	ar_reversal	8,000	M	C	C	C	C	C	I	C	A
12	unequal_variances	8,000	M	I	I	I	I	I	I	C	A
13	mixed_distributions	8,000	M	I	I	I	I	I	I	C	A
14	mcar_reversal	8,000	M	I	I	I	I	I	I	C	A
15	selective_null	8,000	M	C	C	C	C	I	C	C	A
16	bounded_reversal	8,000	P	I	I	I	I	I	I	C	E
17	bounded_reversal_n400	4,000	P	I	I	I	I	I	I	I	E

A.5 Numerical boundary case and substantive limitations

The saved scenario 11, replicate 6628, comparison index 47 has a screened family of 43 entries. At rank 43 the stored p-value is 0.05. The original port evaluates $0.05 \times 43/43$ as 0.049999999999999996 and does not reject; the actual source evaluates $(43/43) \times 0.05$ as 0.05 and rejects. The mathematical bootstrap-grid comparison is $1/20 \leq 1/20$ and rejects. The old referee still reproduces its mismatch on this preserved record; the recovery contracts explicitly distinguish all three identities.

The reuse audit found 31 port-versus-source differences among the original 112,275 studies. At final aggregation, the largest difference in endpoint means between those implementations is 0.00025. Only the exact screened-BH addition changes final endpoint means relative to its original counterpart, with maximum change 8.56×10^{-7}; the other seven exact variants' endpoint means match their originals. Aggregate equality does not establish identity of every rejection vector.

These small arithmetic differences do not repair the larger statistical problems. Full-family historical BH has practical FDR 0.51875 in the block null and 0.778625 in the AR null. Grouped-bootstrap BY has ordinary coverage 0.88775 in the IID equal null and 0.759875 in the AR null. Under AR, cluster-t coverage is 0.8600 in the null and 0.864375 in the reversal scenario because fixed five-window clusters remain dependent. Conversely, low FDR can coexist with negligible power: in the block-reversal scenario, cluster-t practical power is approximately 0.00000658, oracle-normal 0.000296, and grouped-bootstrap zero. Bounded Hoeffding power is zero at $n = 40$ and 0.052632 at $n = 400$, with original-family

bounds approximately $[0, 0.00401]$ and $[0.04462, 0.06065]$. Inconclusiveness, abstention, and conservatism remain findings rather than reasons to select a new reporting threshold.

B Evidence Identities and Reproduction Scope

The C01 full run is `c01-full-167b9074e5711420`; the recovery run is `stat01-recovery-v2-061933a64e27da20`. Execution commits are `2d8342e699f35044044ac518c0b75a3c4a1c768c` and `1ac4c8dd1421aabad05e6ce243cb911a6ed6c8e1`, respectively. The independent evidence check is pinned at commit `6b2eedd7aa254986b85c1bb901587641cbeac8b0`; the supporting analysis at `d22c7751af9c563d5a4d8a304c007f26112c4d12`. The canonical frozen-window hash begins `0d1aa06ccbee`; the physical window-file hash begins `97fbaf6a4b9b`; absolute C01 metrics begin `5700c4bbd738`. Full hashes, source paths, and stored-copy hashes are supplied in `evidence/input-hashes.json` and the input manifest.

The completed C01 campaign performed exhaustive event verification during execution. The final review separately checked all 5,760 terminal records and recomputed their metrics, all 164 launch-file fingerprints, all 144 reused identities, and all 720 request-condition records. Its fresh event replay covered 24 new-cell journals, structurally chosen from the first nonpilot window per source at HIGH_PRESSURE, both capacities, and all adaptations. Independent report checking covers 5,760 absolute rows, 15,840 rankings, 95,040 matched differences, and 26,136 reversal groups. The reproduction scripts regenerate displays only from those saved summaries and verify their fingerprints.

STAT01's recovery validated all 5,360 batches during execution. The final review checked the complete batch inventory and independently recombined all final endpoint moments with high-precision arithmetic; its fresh record-level check covered 200 saved studies across eight scenarios, including the original failing record. It did not claim to rerun all raw studies. Recovery preserves original bytes and labels post-diagnosis comparisons separately. Full raw-record validation and display regeneration are different reproducibility tasks. The manuscript packet supplies the latter, and the complete evidence archive contains the saved raw records with verification recipes; neither workflow generates new outcomes.

The separate budget sensitivity uses the same pinned C01 engine and referee. Its prospective protocol, exact condition identities, source and runner hashes, per-cell outcomes, and independent analysis are retained in `budget-sensitivity/`. Its run hash begins `f19883ccf2c4`. The 12 pilot cells are reused in the 72-cell total, not independent replicates. A documented JSON resume-metadata correction changes no policy, engine, input or configuration. Full new journals are retained in the complete evidence archive. The subsequent `urgency-analysis/` audit reconstructs these 18 SLO-gated histories and checks the finite-work certificate using separate event-ledger and per-request-interval implementations. It executes no new scientific cells; supplied-state predicate tests are engineering checks.

The historical evidence remains under `paper/tompecs-m2`, including its statistical audit, claim changes, cross-metric set comparisons, and finite-reference ranking-recovery analyses. Those materials remain archived support rather than additional independent experiments. Related headroom and module-intervention projects share some infrastructure and topics, but their causal or native-engine results are not used as evidence for this paper. Historical windows, input reconstruction, and the archived 36/990 rule result are reused explicitly; C01 has separate execution semantics and STAT01 separate synthetic truth. Exact overlap with some older cross-project BurstGPT selections remains unresolved. This uncertainty limits claims of independent evidence across projects, not the recorded within-C01 nonoverlap check.

B.1 Timing and ablation evidence

TIME01 and ABL01 execution sources are pinned at `f0d9e525fa99c960129a999993e91bd7a554e997` and `f0142a669488fe184b175828a922128d63c1bc03`. Independent checking recomputed both studies' complete numerical summaries

and replayed all 384 TIME01 journals. For ABL01 it verified all 1,440 result pairs and the one-expression intervention, reviewed the conditional coupling, and replayed 36 controls with their 72 source journals, covering every source–setting–capacity stratum. The remaining controls rely on that coupling and frozen full verification; all journal bytes are checked at packaging. The eight frozen ABL01 analysis tables reproduce byte-for-byte. Complete source manifests, comparison families, rounding diagnostics, and all-source effects accompany the artifact. Reused rows and dependent summaries do not enlarge the independent evidence base.

B.2 Reachable-state proof provenance

The proof integrated in Section 3.4 is preserved with its implementation-assumption checks at commit `32426114d7a321de790a194c51aacbc23703975a`. It leaves the authoritative engine and both policies byte-identical to C01-FULL. Seven hand-derived fixtures and 36 reused budget journals (37,034 checked boundaries) support conformance. This is a proof-first result, not an exhaustive state search or machine-checked formal proof. The proof explicitly excludes arbitrary policy switching and supplied mid-flight states; finite fixture agreement does not extend its scope.

C Chronological Summaries

Figure 4 reports SLO-minus-chunked ANWG differences in four consecutive ten-window blocks. Across settings and capacities, block means range from −0.0120 to +0.2030 for Azure, −0.0300 to +0.2330 for Bailian, and +0.0150 to +0.3275 for BurstGPT. Negative differences occur in 8/48 Azure and 24/48 Bailian block/setting/capacity combinations, and none of the 48 BurstGPT combinations. The frozen EARLY/MIDDLE/LATE summaries show the same qualitative contrast: negative differences in 6/36, 20/36, and 0/36 combinations, respectively. These are corpus-order sensitivities, not estimates of a temporal process or guarantees for future traffic.

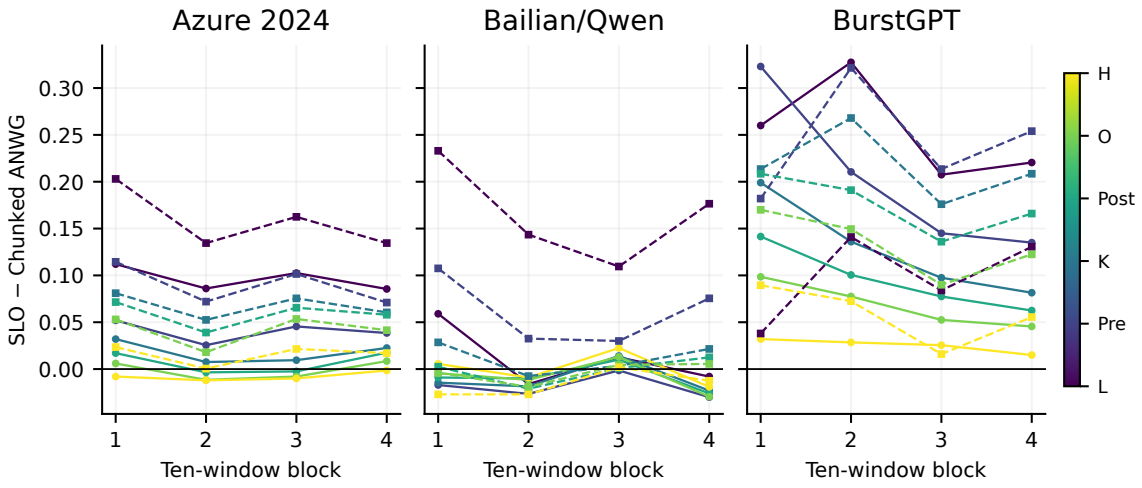


Fig. 4. Chronological block means of SLO-gated minus chunked-FCFS ANWG. Each block contains ten frozen windows. Colors identify inherited settings (L through H); solid circles and dashed squares denote four and eight slots. Blocks have unequal durations and are not independent temporal samples. The figure uses all 144 block/setting/capacity combinations.